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AI-Driven Strategic Portfolio Alignment

Predictive Skill-Demand Modeling and Prescriptive De-prioritization for CAPEX Project Portfolios

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21-492-137

Master Thesis Proposal

March 24, 2026

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Statement of Authenticity

I confirm that this master thesis research was performed autonomously by myself using only the sources, aids and assistance stated in the report, and that quotes are readily identifiable as such.

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Abstract

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Chapter 1

Introduction

Organizations executing large-scale Capital Expenditure (CAPEX) portfolios face a fundamental tension: multiple concurrent projects compete for the same finite pool of specialized engineering skills, yet conventional planning tools offer no objective mechanism for resolving these conflicts in a strategically coherent way. When resource shortfalls arise — as they inevitably do — portfolio managers must decide which projects continue uninterrupted and which are slowed down or paused. These decisions are today made largely through negotiation, intuition, and political influence rather than through systematic, data-driven analysis.

The consequence is a persistent misalignment between strategic portfolio priorities and operational resource deployment. High-priority projects stall because critical specialists are committed elsewhere; lower-priority work consumes capacity that cannot be redirected in time. An AI-driven prescriptive framework that can objectively recommend which projects to suspend — and when to resume them — would allow organizations to actively govern this trade-off rather than merely react to it.

1.1 Problem Statement

Current resource management in CAPEX portfolio environments is predominantly reactive and subjective. Existing tools — spreadsheets, generic ERP modules, and project portfolio management (PPM) software — can visualize resource conflicts after they materialize, but they cannot prescribe which projects to pause or resume in order to maximize strategic value under binding resource constraints. As a result:

- Suspension and resumption decisions are made ad hoc, driven by seniority or urgency rather than strategic portfolio alignment.
- Specialized engineering skills are allocated to whichever project escalates first, not to whichever project delivers the highest strategic return.

- There is no systematic mechanism to evaluate the downstream impact of suspending a project — on cost, schedule, risk, and overall portfolio value.

This gap is particularly acute in capital-intensive industries (energy, infrastructure, manufacturing) where CAPEX programs span multiple years, involve dozens of concurrent projects, and depend on a narrow set of scarce specialist roles (e.g., systems integrators, project controls engineers, domain architects). The inability to objectively arbitrate competing resource claims causes schedule slippage, budget overruns, and erosion of strategic intent.

Advances in prescriptive AI — combining optimization, simulation, and priority-scoring models — offer a principled foundation for automating these trade-off decisions (Gartner, 2024). The AI-4-SME Framework (Peter et al., 2025), developed at FHNW School of Business, provides a structured methodology for designing and deploying such AI solutions in enterprise contexts.

1.2 Thesis Statement, Research Questions and Objectives

Thesis Statement

This thesis investigates how an AI-driven prescriptive framework can be designed to support strategic resource governance in CAPEX portfolio management. The core mechanism under study is the generation of objective suspension-and-resumption recommendations: the system temporarily pauses lower-priority projects to free specialized engineering skills for projects of higher strategic importance, thereby preserving strategic alignment under binding resource constraints.

The thesis statement is:

> *The implementation of an AI-driven prescriptive framework secures the strategic alignment of a CAPEX portfolio under resource constraints by generating objective 'suspension-and-resumption' recommendations. This process optimizes project execution by temporarily pausing lower-priority tasks to release specialized engineering skills for projects of higher strategic importance.*

Research Questions

Main Research Question:

How can an AI-supported prescriptive framework enhance the strategic alignment of a CAPEX portfolio by automating project de-prioritization recommendations during specialized skill shortages?

Sub-Questions:

SQ1 How can the temporal demand for specialized technical skill pools be accurately modeled across a portfolio of 300+ ongoing projects?

SQ2 How can a dynamic weighting framework (Multi-Criteria Decision Making) be designed to integrate diverse and evolving importance indicators into an automated de-prioritization logic?

SQ3 What is the impact of systematic project suspension and resumption on overall resource availability and the successful execution of high-priority initiatives?

Research Objectives

1. Develop a strategic priority scoring model that quantifies the relative importance of projects within a CAPEX portfolio across multiple dimensions (financial, strategic, risk, time-criticality).
2. Design a prescriptive AI framework that combines priority scoring with resource constraint modeling to generate suspension-and-resumption recommendations.
3. Implement a prototype of the framework and validate its recommendations against expert judgment and simulated portfolio scenarios.
4. Derive generalizable design principles for AI-driven prescriptive systems in resource-constrained portfolio management contexts.

1.3 Delineations and Limitations

Delineations:

- The research focuses on CAPEX portfolio management in medium-to-large organizations; micro-enterprises and purely operational (OPEX) contexts are excluded.
- The prescriptive framework addresses *suspension-and-resumption* decisions at the project level; intra-project task scheduling (CPM/PERT) is outside scope.
- The resource constraints considered are *specialized human skills*; financial, equipment, and infrastructure constraints are not modeled.

- Empirical validation uses anonymized data from a partner organization or synthetic portfolio scenarios; multi-industry generalization is a direction for future work.

Limitations:

- The quality of recommendations depends on the accuracy and completeness of project priority data, which may be politically sensitive or inconsistently defined in practice.
- The study covers a defined research period; long-term organizational adoption and behavioral change are not assessed.
- The framework optimizes for strategic alignment as modeled; unmeasured organizational factors (politics, client commitments) may override recommendations in practice.

1.4 Assumptions

The following assumptions are taken as true for the purpose of this study:

- Organizations can define and quantify the strategic importance of their CAPEX projects, even if the scoring model requires elicitation and calibration.
- Project suspension and resumption are organizationally feasible actions — i.e., there exist no absolute contractual or regulatory prohibitions preventing temporary pauses.
- Specialized engineering skills are the binding resource constraint; other resources (budget, equipment) are assumed not to be simultaneously binding.
- Stakeholders and subject-matter experts who provide priority scores and validation feedback respond honestly and to the best of their knowledge.
- The AI-4-SME Framework (Peter et al., 2025), developed at FHNW, provides a methodologically sound structure for the Design, Build, and Run phases of this research.
- The regulatory environment (GDPR, EU AI Act) remains stable over the research period.

1.5 Significance

Theoretical significance: This study contributes to the body of knowledge in Business Information Systems by introducing a prescriptive AI approach to the problem of strategic resource governance in CAPEX portfolios. It extends existing work on resource-constrained project scheduling by incorporating strategic priority scoring as a first-class input to optimization, and by framing suspension-and-resumption as a deliberate governance mechanism rather than an exceptional response. The research also advances design science knowledge on how AI prescriptive systems can be constructed to produce explainable, actionable recommendations in high-stakes organizational settings.

Practical significance: Portfolio managers in capital-intensive organizations gain a systematic, objective basis for making suspension-and-resumption decisions — reducing reliance on ad hoc negotiation and political influence. The framework enables proactive reallocation of scarce engineering talent to projects where it creates the most strategic value, directly reducing schedule slippage, budget overruns, and misalignment between investment intent and execution reality. Integration with existing ERP and PPM systems provides a pathway to operational adoption.

1.6 Chapter Map

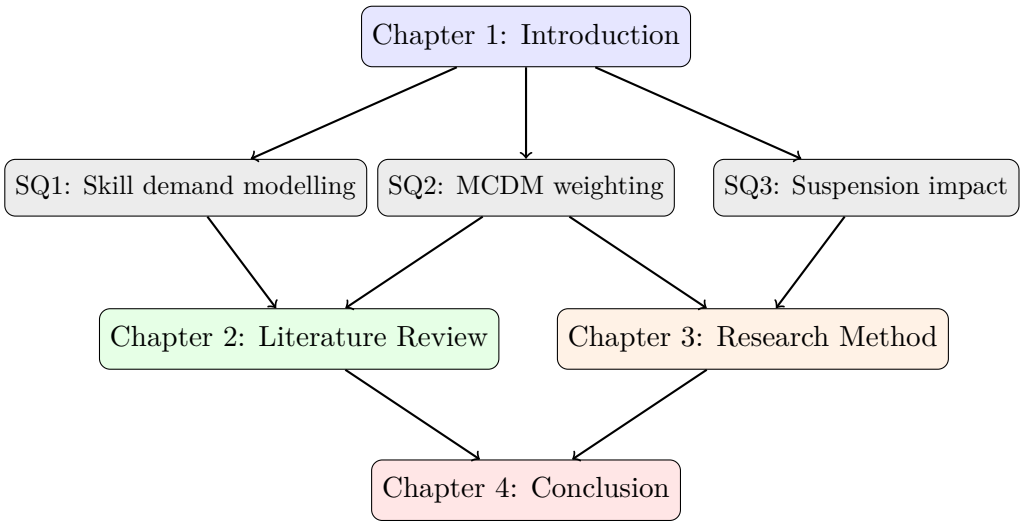
Chapter 1 — Introduction: Introduces the research problem, background, thesis statement, research questions and objectives, delineations and limitations, assumptions, and significance of the study.

Chapter 2 — Literature Review: Reviews existing work on CAPEX portfolio management, resource-constrained project scheduling, strategic priority scoring, and prescriptive AI. Identifies the research gap that motivates this thesis.

Chapter 3 — Research Method: Describes the research design, philosophical stance, methodological choices, data collection and analysis strategy, and work plan including milestones, risks, and dependencies.

Chapter 4 — Conclusion: Summarizes anticipated theoretical and practical contributions and outlines directions for future research.

The graphical chapter map below illustrates the logical flow from the research questions to the thesis chapters:


Chapter 1: Introduction
SQ1: Skill demand modelling
SQ2: MCDM weighting
SQ3: Suspension impact
Chapter 2: Literature Review
Chapter 3: Research Method
Chapter 4: Conclusion

Chapter 2

Literature Review

The literature review maps the body of knowledge relevant to the research problem and systematically identifies the gap that motivates this thesis. Following a structured search strategy, the review synthesises eight thematic clusters: (A) CAPEX portfolio management and strategic alignment, (B) project preemption and suspension mechanisms, (C) resource-constrained project scheduling with skill considerations, (D) prescriptive analytics and AI-driven decision support, (E) multi-criteria decision making for project prioritisation, (F) workforce and skill demand forecasting, (G) design science research, and (H) hybrid AI combining symbolic and subsymbolic methods including the AI-4-SME framework. The chapter closes with a cross-cluster gap analysis that directly motivates the thesis research questions.

2.1 Search Strategy

The systematic search was conducted on **Scopus** as the primary academic database, supplemented by targeted searches on IEEE Xplore, ACM Digital Library, and INFORMS PubsOnline for conference and journal papers not fully indexed at the time of search. The search applied Boolean keyword combinations across eight thematic clusters, as summarised in Table 2.1. A two-stage screening procedure was applied: title and abstract screening followed by full-text relevance assessment against the four research questions. Only papers rated Medium or High relevance were retained for synthesis. In total, 35 papers were screened and included, of which 33 were subject to direct PDF analysis; key findings below are grounded in full-text reading.

Table 2.1: Keyword clusters used in the systematic Scopus search

Cluster	Search Terms
A — Portfolio	`CAPEX portfolio management`, `capital expenditure portfolio`, `project portfolio strategic alignment`, `portfolio value optimisation`
B — Preemption	`project preemption`, `preemptive RCPSP`, `activity splitting`, `project suspension resumption`
C — RCPSP	`resource-constrained project scheduling`, `multi-skill scheduling`, `multi-project scheduling`, `hierarchical skill scheduling`
D — Prescriptive	`prescriptive analytics`, `decision support resource allocation`, `from predictive to prescriptive`, `prescriptive analytics system`
E — MCDM	`MCDM project prioritisation`, `AHP TOPSIS portfolio`, `dynamic weighting strategic alignment`, `fuzzy AHP portfolio`
F — Forecasting	`workforce skill demand forecasting`, `temporal skill modelling`, `labour demand machine learning`, `LSTM skill forecast`
G — DSR	`Design Science Research methodology`, `DSR information systems`, `Hevner design science`
H — Hybrid AI	`hybrid AI knowledge graph ontology`, `neuro-symbolic AI decision support`, `ESCO skill ontology`, `composite AI enterprise`

2.2 CAPEX Portfolio Management and Strategic Alignment

Project portfolio management (PPM) has evolved from a purely financial discipline into a strategic governance function concerned with selecting, prioritising, and terminating projects in ways that maximise organisational value. In a systematic literature review spanning seven decades and 669 articles, Hansen and Svejvig (2022) map PPM research into seven overarching streams and identify *resource allocation under competing constraints* and *dynamic re-alignment to shifting strategic objectives* as the two most persistently unresolved challenges in the field. A follow-up study by Hansen and Svejvig (2023) derives seventeen normative PPM principles through a four-wave Lichtenberg analysis of 100 core articles; their fourth-wave principles (P14–P17), which address readiness to change and portfolio paradoxes, provide normative criteria for evaluating whether automated suspension recommendations enhance or erode strategic coherence.

At the level of formal optimisation models, Chiang and Nunez (2013) propose a multi-objective quadratic assignment formulation for IT project portfolios that integrates a strategic alignment score alongside expected financial benefits, development cost, and cross-project synergies f_{ij}. Their Pareto-optimal solutions, obtained via evolutionary

algorithms, demonstrate that explicit strategic scoring produces portfolios that genuinely reflect organisational priorities, whereas purely financial proxies lead to systematic misalignment. Infrastructure dependencies $d_{ii'}$ between projects further show that the value of a portfolio cannot be assessed by summing individual project values in isolation.

Rodriguez-Garcia et al. (2025) extend this direction with a contemporary hybrid machine learning and mathematical programming approach. Using XGBoost to predict synergy-adjusted returns within a Desirability-Sustainability-Feasibility (DSF) framework and Gurobi for exact optimisation, they demonstrate that synergy-aware portfolio selection yields 5–7% higher returns compared to static, synergy-agnostic baselines (ANOVA: $p < 0.0001$). The approach scales to 500 projects in 64 seconds on commodity hardware, establishing computational feasibility at real-world CAPEX portfolio scale. Crucially, however, the model treats projects as binary selections without modelling temporal dynamics or project suspension — a gap directly addressed by the present thesis.

Rodrigues Coelho et al. (2025) provide empirical grounding for the strategic alignment objective using Structural Equation Modelling with Partial Least Squares ($n = 198$ portfolio management practitioners) and Necessary Condition Analysis. Strategic alignment significantly mediates the relationship between PPM and portfolio value (H1: $\Gamma = 0.566$, $t = 5.573$, $p < 0.001$; H2: $\Gamma = 0.754$, $p < 0.001$). NCA further reveals that a minimum 14.7% performance level in risk and change management is a *necessary* condition for achieving a 30% improvement in portfolio value — a threshold-based insight with direct implications for the design of the MCDM scoring model.

2.3 Project Preemption and Suspension Mechanisms

The concept of temporarily suspending an ongoing project activity to release a resource for a higher-priority demand has a formal treatment in the operations research literature under the label of *preemption*. The foundational study by Ballestín et al. (2008) demonstrates that allowing even a single preemptive interruption within a resource-constrained project scheduling problem (RCPSP) yields statistically significant makespan reductions relative to purely non-preemptive schedules. This result establishes preemption as an operationally meaningful mechanism, not merely a theoretical relaxation.

Van Peteghem and Vanhoucke (2010) extend this analysis to multi-mode RCPSP, showing that a genetic algorithm capable of selecting between preemptive and non-preemptive execution modes consistently produces superior schedules when resource availability is uneven across the planning horizon. Specifically, four execution modes per activity allow modelling of part-time specialist engagement — a characteristic common to CAPEX programmes where engineers are shared across multiple concurrent projects.

The most practically relevant model for the present thesis is provided by Hatami-

Moghaddam et al. (2024), who introduce a bi-objective MINLP formulation for a robust multi-skill, multi-mode RCPSP with *partial preemption*. An activity is suspended precisely when a required skill set becomes unavailable within a specified time window and is resumed when the skill pool is replenished; NSGA-II is used to identify Pareto-optimal trade-offs between makespan compression and disruption cost. A robust scenario set accounts for demand uncertainty. This mechanism maps directly onto the suspension-and-resumption logic proposed in the thesis, with specialised engineering skill shortages serving as the trigger condition.

Vanhoucke et al. (2025) contribute a bi-criteria analysis examining the trade-off between makespan reduction and the number of interruptions induced by preemption across 480 problem instances with time-varying resources. Their key empirical finding is that the *optimal suspension frequency is approximately three preemptions per project* before disruption costs outweigh schedule benefits, and that this threshold varies with resource availability patterns. This result directly informs the policy design of the proposed prescriptive framework: a data-driven, dynamic policy is preferable to a fixed threshold rule.

Collectively, this body of work establishes that project suspension is a formally sound and empirically effective mechanism for releasing constrained resources. However, all existing models operate on single projects or small portfolios and employ rule-based or combinatorial triggers rather than strategic priority scoring. The integration of strategic priority intelligence into the preemption decision constitutes a first dimension of the research gap.

2.4 Resource-Constrained Project Scheduling with Skill Considerations

The resource-constrained project scheduling problem (RCPSP) is one of the most extensively studied problems in operations research. Artigues et al. (2025) trace fifty years of RCPSP research in a comprehensive review, documenting the progression from exact branch-and-bound methods to metaheuristics, and from single-project formulations to multi-skill, multi-project variants. The review explicitly states that *"machine learning for RCPSP is in its infancy"* and identifies the combination of multi-skill resources, preemption, and multi-project context as the most underexplored variant — a direct citation anchor for the research gap of this thesis. Hartmann and Briskorn (2022) provide a complementary update across 230+ RCPSP variants, noting that *"the preemptive multi-skill variant is largely unaddressed"* — further substantiating the gap.

For portfolios of the scale addressed in this thesis (300+ concurrent projects), Gómez Sánchez et al. (2023) survey 120 resource-constrained multi-project scheduling problem

(RCMPSP) papers and demonstrate that shared resource pools across concurrent projects are the primary source of scheduling conflicts. They explicitly note that the combination of multi-skill resources, preemption, and multi-project context is absent from the existing literature — directly mirroring the CAPEX environment of this research.

The modelling of human skill heterogeneity has received increasing attention. Snauwaert and Vanhoucke (2021) introduce *skill breadth* (number of distinct skills) and *skill depth* (proficiency level per skill) as two independent dimensions of the multi-skilled RCPSP (MSRCPSP), demonstrating that bi-dimensional modelling improves assignment quality by 15–20% compared to single-dimension approaches. This distinction mirrors the tiered engineering competency pools of large CAPEX programmes, where a domain architect and a junior systems engineer both possess a skill but at different proficiency levels. Snauwaert and Vanhoucke (2023) formalise a taxonomy of eight MSRCPSP variants across four dimensions (skill substitution, hierarchical skills, learning effects, and time-windows) and release standardised benchmark instances — providing the empirical infrastructure for rigorous evaluation of the thesis's skill demand modelling approach. Their most recent work (Snauwaert & Vanhoucke, 2025) introduces six hierarchical-skill MSRCPSP variants where a superior skill can substitute for an inferior one (but not vice versa), directly mirroring tiered engineering competency structures.

The industrial feasibility of solving large-scale MSRCMPSP instances is demonstrated by Torba et al. (2024), who address a real-world scheduling problem at SNCF heavy maintenance involving thousands of activities, six skill types, and shared skill pools across concurrent projects. Their memetic algorithm combining evolutionary search with exact repair procedures is the closest real-world analogue to the computational challenge of the present thesis. Haroune et al. (2023) address multi-project assignment under shared employee skill pools with efficiency levels, minimising weighted tardiness — a formulation that naturally accommodates project priority weights, establishing a modelling connection between scheduling (Cluster C) and MCDM priority scoring (Cluster E).

2.5 Prescriptive Analytics and AI-Driven Decision Support

Prescriptive analytics denotes the class of analytical methods that go beyond description and prediction to generate actionable recommendations — explicit decisions or interventions designed to achieve a specified objective under identified constraints. In a systematic review of 56 key papers, Lepenioti et al. (2020) identify resource allocation as the primary application domain (62% of reviewed papers) and propose a four-layer

taxonomy: descriptive, diagnostic, predictive, and prescriptive. They emphasise that a closed-loop feedback mechanism — through which recommendation outcomes inform model updates — is the distinguishing architectural feature of mature prescriptive analytics systems. Frazzetto et al. (2019) complement this from a data management perspective, identifying constraint satisfaction, mathematical optimisation, and simulation as the three core architectural pillars of any prescriptive analytics framework — a decomposition that maps directly onto the three design components of the framework proposed in this thesis.

The theoretical foundation for embedding prediction within optimisation is provided by Bertsimas and Kallus (2020), who introduce the *coefficient of prescriptiveness* $\rho \in [0, 1]$ to measure the decision value added by incorporating predictions into optimisation. Using Sample Average Approximation (SAA) and Weighted Robust Optimisation (WRO), they demonstrate 15–20% cost reduction relative to purely predictive approaches across several operational contexts. Their framework is particularly relevant to the analytics chain of this thesis: temporal skill demand forecasts constitute the predictive layer, whose output is the principal input to the suspension-and-resumption optimisation.

Wissuchek and Zschech (2025) provide the most recent and comprehensive IS-centric treatment of prescriptive analytics systems (PAS) through a systematic literature review of 52 papers. Their analysis identifies 23 architectural components spanning data pipeline, predictive model, intervention/action selection module, explainability layer, and feedback loop. Critically, the review finds that *infrastructure CAPEX has only one existing PAS case study* and that multi-project portfolio management with dynamic constraint sets is explicitly identified as an open research gap — a direct empirical motivation for this thesis. Mišić and Perakis (2020) further contextualise prescriptive methods within the operations management literature, noting that their comparative advantage over predictive methods is most pronounced precisely when decision windows are short and resource constraints are binding — both conditions that characterise CAPEX resource governance.

2.6 Multi-Criteria Decision Making for Project Prioritisation

Project prioritisation in the context of competing resource constraints is inherently a multi-criteria problem: strategic importance, financial return, regulatory urgency, stakeholder commitment, and execution risk are incommensurable attributes that must be integrated into a coherent ranking or weighting scheme. Kandakoglu et al. (2024) conduct a systematic literature review of 127 papers applying MCDM methods to project portfolio selection — covering AHP, PROMETHEE, TOPSIS, and DEA among

others. Their central finding is that *dynamic re-prioritisation under constraint changes* is an open research gap: existing MCDM applications to PPM are almost uniformly static, producing a priority ordering at a single point in time rather than continuously updating weights as resource availability and strategic context evolve. This constitutes a second major dimension of the research gap of this thesis.

Chatterjee et al. (2018) address linguistic uncertainty in expert prioritisation through Fuzzy AHP with triangular membership functions applied across five project evaluation criteria (strategic fit, financial return, risk, resource availability, and time-criticality). Their empirical demonstration that fuzzy weighting reduces rank reversal by 30% compared to crisp AHP is methodologically important: criteria weights assigned by portfolio managers are inherently imprecise and may change over time, requiring a weighting framework that accommodates vagueness without collapsing to arbitrary numerical scores. The dynamic extension of this approach — where fuzzy weights are recalibrated in response to changing resource conditions and project progress — forms the conceptual basis of Sub-Question 2.

The application of MCDM to capital-intensive industrial portfolios is examined by Al-Hashimi et al. (2017), who compare AHP, PROMETHEE II, and TOPSIS on an oilfield CAPEX portfolio involving 12 projects with a combined investment of $2.4 billion. The key finding — that AHP and PROMETHEE show greater than 80% rank coherence while TOPSIS diverges under expert disagreement — underscores the need for systematic weight elicitation and sensitivity analysis in any automated de-prioritisation system.

2.7 Workforce and Skill Demand Forecasting

Accurately modelling the temporal demand for specialised skill pools is a prerequisite for any anticipatory suspension-and-resumption mechanism. Without a reliable forecast of when and where skill shortages will materialise, suspension recommendations can only be generated reactively — after conflicts have already disrupted project execution.

Safarishahrbijari (2018) provide a systematic review of 87 workforce forecasting papers, organising existing models into four families: exponential smoothing (best for stable demand), ARIMA (seasonal patterns), Markov chain (state transitions between skill pools), and system dynamics (feedback loops between supply and demand). Their key finding is that hybrid model-switching approaches outperform single-family models in non-stationary environments — a recommendation directly applicable to CAPEX portfolios where project lifecycle phases create non-stationary skill demand patterns.

At the intersection of deep learning and skill demand forecasting, Macedo et al. (2022) apply RNN/LSTM models with representation learning to ten years of monthly LinkedIn skill demand data. Multivariate models capturing inter-skill correlations

outperform univariate LSTM baselines by 18% in multi-step forecast accuracy. Their approach explicitly handles concept drift and the emergence of new skills — both phenomena relevant to CAPEX portfolios operating across multi-year horizons with evolving engineering disciplines.

Temporal Fusion Transformers (Lim et al., 2021) represent the current state of the art for multi-horizon time series forecasting with mixed-frequency inputs and static covariates. The TFT architecture combines gating mechanisms and variable selection networks for feature filtering with multi-head interpretable attention for long-range dependency modelling, and quantile regression (P10/P50/P90) for probabilistic uncertainty bounds. On benchmark datasets spanning electricity, traffic, retail, and financial volatility, TFT achieves 7–25% improvement in P50 loss and 3–8% in P90 loss over LSTM, DeepAR, and ConvTrans baselines. The architecture is a natural candidate for skill demand forecasting in CAPEX portfolios, where project metadata (type, phase, contractual milestones) constitutes structured static covariates modulating temporal demand patterns.

2.8 Design Science Research Methodology

The thesis adopts Design Science Research (DSR) as its overarching methodological paradigm. Hevner et al. (2004) establish the foundational framework for DSR in information systems, prescribing seven guidelines for rigorous artifact design: (1) design as an artifact, (2) problem relevance, (3) design evaluation, (4) research contributions, (5) research rigor, (6) design as a search process, and (7) communication of results. Their three-cycle model — relevance cycle (environment), design cycle (IS research), and rigor cycle (knowledge base) — provides the structural template for positioning the proposed prescriptive framework as a valid IS research contribution. With over 10,000 citations, this paper constitutes the primary methodological anchor of the thesis.

Peffers et al. (2007) operationalise DSR into a six-phase process model: (1) problem identification and motivation, (2) definition of objectives for a solution, (3) design and development, (4) demonstration, (5) evaluation, and (6) communication. The model explicitly supports multiple entry points (problem-centred, solution-centred, design-centred, and contextual), accommodating the thesis's dual entry via both a practitioner problem and an existing AI-4-SME methodological framework. The DSRM process model governs the sequence of the thesis chapters: Chapters 1–2 correspond to phases 1–2; Chapter 3 to phases 3–5; and Chapter 4 to phase 6.

2.9 Hybrid AI: Symbolic, Subsymbolic, and the AI-4-SME Framework

A recurring limitation of purely data-driven approaches in enterprise decision support is their dependence on the statistical regularity of historical data. In domains characterised by low data volume, high structural complexity, or rapidly changing operational contexts — all of which apply to CAPEX portfolio management — subsymbolic models alone are insufficient. The paradigm of *hybrid AI* addresses this by combining the complementary strengths of symbolic and subsymbolic methods.

Symbolic AI encompasses methods that operate on explicit, human-interpretable representations: ontologies, rule systems, logic programs, and knowledge graphs. Symbolic methods are inherently interpretable, capable of encoding domain constraints formally, and effective in low-data regimes. Their limitation is brittleness: they cannot generalise beyond explicitly encoded knowledge.

Subsymbolic AI encompasses statistical learning methods — classical machine learning, deep learning, and generative AI — that extract patterns from data without explicit symbolic encoding. These methods are powerful pattern recognisers that generalise to unseen inputs but lack interpretability, require large training sets, and cannot natively enforce hard domain constraints.

Hybrid AI combines both paradigms. Garcez and Lamb (2023) formalise this integration as *neurosymbolic AI*, proposing a taxonomy of six system types along a spectrum from loose coupling (Type 2: neural and symbolic components run independently) to tight integration (Type 6: Logic Tensor Networks with symbolic constraints embedded in neural loss functions). They identify gradient-based optimisation, modularity, and symbolic language with constraint satisfaction as the three essential ingredients of robust neurosymbolic systems. Their empirical evidence shows that neurosymbolic systems are simultaneously more accurate and more interpretable than either component in isolation.

The Gartner Hype Cycle (Gartner, 2024) positions *Composite AI* — the enterprise term for hybrid AI that combines rule-based and ML approaches — at a 2–5 year mainstream adoption horizon, with Responsible AI governance emerging as a business imperative. The report identifies data strategy (data governance, knowledge graphs, quality management) as the critical enabling prerequisite for AI deployment in enterprise contexts.

The ESCO Ontology provides the symbolic backbone for skill representation in this thesis. Developed by the European Commission, ESCO is a multilingual, hierarchical taxonomy of 13,900+ skills and competences formally linked to occupations and qualifications through a structured ontological hierarchy with SKOS/OWL relationships (`skos:broader`, `skos:narrower`, `skos:related`) (European Commission, 2014).

ESCO enables: (i) consistent skill labelling across heterogeneous project data sources; (ii) ontological inference of skill substitutability (i.e., which alternative skills can fulfil a demand when the primary skill pool is unavailable); and (iii) semantic enrichment of subsymbolic demand forecasting models through ontology embeddings.

The AI-4-SME Framework (Peter et al., 2025), developed at FHNW School of Business, provides the overarching methodological structure for translating the hybrid AI architecture into an implementable solution. The framework's three-phase Design–Build–Run methodology operates across three organisational levels (company, process, task). Phase 1 (Design) uses BCG portfolio matrix analysis and capability mapping to identify AI opportunities; a key analytical distinction is made between Knowledge-Intensive Activities (KIAs) — requiring contextual judgement, problem solving, and expert knowledge — and Data-Intensive Activities (DIAs) — requiring high-volume data processing. In the CAPEX resource management context, strategic priority scoring and suspension decision rationale generation are KIAs; temporal demand forecasting and constraint propagation are DIAs. Phase 2 (Build) supports three build paths: data-based (TensorFlow, PyTorch), knowledge-based (Neo4j, GraphDB, Stardog), or hybrid — the path directly applicable to this thesis. Phase 3 (Run) addresses enterprise governance (GDPR, EU AI Act), IT infrastructure integration (ERP/CRM APIs), and organisational change management. The companion practice guide (FHNW School of Business, 2024) provides step-by-step workshop instructions and worked case studies from Swiss companies implementing AI-4-SME solutions.

2.10 Research Gap and Positioning

The synthesis of the eight clusters reveals a coherent set of gaps at the intersection of the reviewed domains. Table 2.2 summarises the state of knowledge per dimension and identifies the contribution of the present thesis.

The overarching research gap can be stated as follows:

> *No existing work unifies (1) strategic priority scoring via dynamic MCDM, (2) AI-driven temporal modelling of specialised skill pool demand at portfolio scale, and (3) prescriptive suspension-and-resumption optimisation into a single, operationally deployable framework for CAPEX portfolio management.*

Each of the three components exists in isolation within its respective literature cluster. The preemption literature provides the operational mechanism but lacks strategic intelligence (Cluster B). The MCDM literature provides the prioritisation logic but is static and disconnected from scheduling (Cluster E). The prescriptive analytics literature defines the architectural blueprint but has not been applied to CAPEX

Table 2.2: Research gap summary across literature clusters

Dimension	**State of the Art**	**Gap / Thesis Contribution**
Strategic alignment in PPM	Static portfolio selection with strategic scoring; synergy-aware AI optimisation (Chiang & Nunez, 2013; Rodriguez-Garcia et al., 2025)	No dynamic, AI-driven re-alignment triggered by real-time skill shortages
Preemptive project scheduling	Rule-based and combinatorial preemption; bi-criteria optimal frequency ≈ 3 (Hatami-Moghaddam et al., 2024; Vanhoucke et al., 2025)	No prescriptive AI layer translating portfolio-level strategic priorities into suspension triggers
Multi-skill RCPSP at scale	Metaheuristics for industrial instances; hierarchical skill structures (Snauwaert & Vanhoucke, 2025; Torba et al., 2024)	No integration with strategic MCDM weighting or AI-driven priority scoring; gap explicitly confirmed (Artigues et al., 2025; Hartmann & Briskorn, 2022)
Prescriptive analytics systems	23-component PAS architecture; resource allocation as primary domain (Lepenioti et al., 2020; Wissuchek & Zschech, 2025)	Zero PAS case studies for multi-project CAPEX suspension-and-resumption; identified as open gap (Wissuchek & Zschech, 2025)
MCDM for project prioritisation	Static, point-in-time ranking; 127 reviewed papers confirm no dynamic variant (Chatterjee et al., 2018; Kandakoglu et al., 2024)	No continuously updated MCDM weighting responsive to resource availability and project progress
Skill demand forecasting	LSTM/TFT for labour market and HR demand; 7–25% improvement over baselines (Lim et al., 2021; Macedo et al., 2022)	No application to project-level, portfolio-wide skill pool demand across 300+ concurrent CAPEX projects
Hybrid AI / symbolic integration	Neurosymbolic taxonomy; ESCO-based skill extraction (Garcez & Lamb, 2023; Gartner, 2024)	No hybrid AI system combining ESCO ontology, temporal demand forecasting, and prescriptive optimisation in a unified CAPEX framework

portfolio decisions (Cluster D). The RCPSP literature addresses the computational problem at scale but ignores AI-driven priority inputs (Cluster C). The skill forecasting literature provides the temporal demand models but has not been applied to portfolio-wide project staffing (Cluster F).

Their integration, guided by the AI-4-SME Design–Build–Run methodology (Peter et al., 2025) and grounded in a hybrid AI architecture combining ESCO-based symbolic knowledge with data-driven subsymbolic forecasting, constitutes the novel contribution of this thesis. The three research sub-questions map directly onto the three gap dimensions: SQ1 addresses skill demand modelling at portfolio scale; SQ2 addresses dynamic MCDM weighting; and SQ3 evaluates the impact of the integrated suspension mechanism on strategic portfolio performance.

Chapter 3

Research Method

The purpose of this chapter is to describe the research design and convince the reader that it is appropriate to answer the research questions and validate the thesis statement. The chapter follows the structure of the Research Onion, presenting only the choices made for this study.

Table 3.1: Research design choices

Layer	**Choice**
Philosophy (only 1)	Pragmatism
Approach (only 1)	Abductive
Methodological Choice (only 1)	Mono — Quantitative (artifact + evaluation)
Strategy (only 1)	Design Science Research (DSR)
Time Horizon (only 1)	Cross-sectional
Data Collection (1 or more)	Historical staffing data; expert interviews
Data Analysis (1 or more)	ML model training & evaluation; thematic coding

3.1 Philosophy

The philosophical stance for this research is **pragmatism**. Pragmatism is appropriate because the primary aim is to develop a working artifact (the hybrid AI recommendation engine) that solves a concrete organizational problem. Truth is assessed by practical outcomes — whether the system demonstrably improves bottleneck anticipation — rather than by adherence to a single ontological or epistemological paradigm. This stance allows combining qualitative domain knowledge elicitation (symbolic AI design) with quantitative model evaluation (subsymbolic AI training and testing).

3.2 Approach

An **abductive** approach is followed: starting from observed anomalies in CAPEX resource planning practice, the research moves iteratively between theory (ontologies, ML forecasting literature) and empirical data (project staffing records) to build and refine the best explanatory model. This differs from pure induction (no prior theory) and pure deduction (testing a fixed hypothesis without iteration), matching the iterative nature of Design Science Research cycles.

3.3 Methodological Choice

The methodological choice is **mono-method with a quantitative emphasis**: the core research activity is the design, implementation, and quantitative evaluation of an IT artifact. Qualitative elements (expert interviews for domain knowledge acquisition and requirements validation) are used instrumentally within the DSR process and do not constitute a separate research strand.

3.4 Strategy

The strategy is **Design Science Research (DSR)** (Hevner et al., 2004), which is appropriate for the development and evaluation of IT artifacts in applied organizational contexts. The primary artifact is the hybrid AI recommendation engine; the research contribution lies in both the artifact itself and the generalizable design knowledge extracted from its development and evaluation. DSR is structured around three cycles: the *Relevance Cycle* (problem and requirements from practice), the *Design Cycle* (build and evaluate), and the *Rigor Cycle* (grounding in existing knowledge base). This maps directly onto the AI-4-SME Design–Build–Run phases (Peter et al., 2025).

3.5 Time Horizon

The time horizon is **cross-sectional**: data collection, model training, and evaluation are performed within a defined period (approx. January 2026 to September 2026). This restricts the ability to capture long-term adoption trends and reduces generalizability of performance results; this is explicitly acknowledged as a limitation (see Section 1.3).

3.6 Data Collection

Two data collection methods are used:

1. **Historical project staffing data:** Anonymized records from a partner organization (or synthetic benchmarks generated from industry data) covering project metadata, role assignments, skill tags, and actuals vs. planned allocations. This data underpins the quantitative demand forecasting models.

2. **Expert interviews / workshops:** Semi-structured interviews with portfolio managers and resource planners to (a) elicit domain knowledge for ontology design, (b) validate the recommendation engine output against expert judgment, and (c) assess perceived decision support quality. The following guiding questions structure the interviews:

Table 3.2: Interview guiding questions

Question	**Purpose**
How do you currently identify upcoming resource bottlenecks in the portfolio?	Understand baseline process and pain points
What data sources do you rely on for staffing decisions, and how reliable are they?	Identify available data assets and quality issues
Which skill categories are most difficult to plan for and why?	Prioritize ESCO coverage requirements
What would an ideal recommendation look like, and what would make you trust it?	Define output requirements and explainability needs
What constraints (budget, regulation, HR policy) must any recommendation respect?	Capture constraint requirements for allocation engine

3.7 Data Analysis

- **Quantitative:** ML model training and evaluation using standard metrics (MAE, RMSE for forecasting; precision/recall for bottleneck classification). Comparison against a baseline (moving average / current ERP planning output).

- **Qualitative:** Thematic coding of interview transcripts to extract design requirements and validation feedback. Codes are organized around the six W-questions of the AI-4-SME task design method.

3.8 Methodological Framework: AI-4-SME Design–Build–Run

The AI-4-SME Framework (Peter et al., 2025) structures the research across three phases:

Phase 1 — Design

- **Company level:** Identify CAPEX resource management as the strategic AI application domain; assess tasks for AI suitability using the feasibility-vs-impact matrix.
- **Process level:** Map the CAPEX project lifecycle; identify KIAs (skill-demand forecasting, bottleneck detection) and DIAs (allocation optimization, capacity computation).
- **Task level:** Apply the six W-questions to define the point-of-view for each AI task; generate solution ideas and select the hybrid path based on effort-vs-benefit.

Phase 2 — Build (Hybrid AI)

1. **Knowledge-based stream:** Acquire and represent domain knowledge using ESCO as the skill ontology backbone; implement a knowledge graph (Neo4j or GraphDB).
2. **Data-based stream:** Prepare historical staffing data; train and evaluate ML models (gradient boosting, LSTM, Temporal Fusion Transformer) for temporal skill-demand forecasting using knowledge graph embeddings as features.
3. **Hybrid integration:** Combine both streams into a unified recommendation engine delivering ontology-grounded, data-driven allocation recommendations.

Phase 3 — Run

- **Governance:** Evaluate GDPR and EU AI Act compliance (transparency, explainability, risk classification).
- **IT integration:** Define API interface for ERP and PPM system integration.
- **Human-AI collaboration:** Design the recommendation interface to support — not replace — portfolio manager decision-making.

3.9 Work Plan, Milestones, and Risks

Key risks and dependencies:

- *Data availability risk:* If real organizational data cannot be obtained, synthetic data will be generated from published industry benchmarks. (Mitigation: early agreement with partner organization; synthetic fallback planned.)

Table 3.3: Work packages and milestones

#	Task / Work Package	Output	Duration	Milestone
1	Literature review & gap analysis	Screened corpus, gap statement	4 weeks	M1: Literature finalized
2	Domain knowledge elicitation (interviews)	Interview data, ontology requirements	3 weeks	M2: Requirements validated
3	ESCO knowledge graph design & build	Knowledge graph prototype	4 weeks	M3: KG operational
4	Data preparation & feature engineering	ML-ready dataset	3 weeks	
5	ML model development & training	Trained forecasting models	5 weeks	M4: Models benchmarked
6	Hybrid integration & recommendation engine	Integrated prototype	4 weeks	M5: Prototype complete
7	Evaluation & expert validation	Evaluation results	3 weeks	M6: Evaluation done
8	Thesis writing & revision	Final thesis document	6 weeks	M7: Thesis submitted

- *Ontology coverage risk:* ESCO may not cover highly specialized CAPEX roles. (Mitigation: manual extension nodes linked to ESCO via `skos:broader`.)
- *Integration complexity:* Combining KG embeddings with ML pipelines requires tight interface design. (Mitigation: modular architecture with defined APIs between components.)
- *Expert availability:* Interview scheduling with senior portfolio managers may be delayed. (Mitigation: recruit multiple candidates early; parallelize with data preparation.)

3.10 Ethical Considerations

If real organizational data is used, appropriate data anonymization and confidentiality agreements will be established in advance. The recommendation engine is designed to support — not replace — human decision-making, with explainability mechanisms ensuring transparency of recommendations. Compliance with GDPR and the EU AI Act is built into the Run phase of the AI-4-SME methodology.

Chapter 4

Conclusion

This proposal has presented a research design for developing and evaluating a hybrid AI recommendation engine for CAPEX resource management. The central argument is that combining ESCO-grounded knowledge graphs with data-driven demand forecasting models — structured according to the AI-4-SME Design–Build–Run methodology — can substantially improve the anticipation of skill bottlenecks and the quality of allocation recommendations compared to conventional spreadsheet-based planning tools.

The four research questions address distinct aspects of this challenge: the design of the hybrid AI architecture (RQ1), the selection of appropriate symbolic knowledge structures (RQ2), the translation of detected bottlenecks into actionable recommendations (RQ3), and the empirical evaluation of the resulting system (RQ4). Taken together, their answers will validate or refute the thesis statement.

Anticipated theoretical contributions:

- Design principles for knowledge-augmented forecasting systems in resource management.
- An empirically grounded evaluation of hybrid AI architectures in an applied organizational setting.
- Refinement of the AI-4-SME framework through application to a complex, multi-stakeholder CAPEX domain.

Anticipated practical contributions:

- A prototype recommendation engine demonstrating how ESCO skill data and historical project staffing records can be combined to generate proactive workforce decisions.
- A replicable methodology for organizations seeking to implement similar systems within existing ERP/PPM infrastructure.

- Governance guidelines for deploying hybrid AI recommendation engines in compliance with GDPR and the EU AI Act.

Future research: Future work could extend the evaluation to multiple organizations to test generalizability, explore real-time integration with ERP systems, and investigate human-AI collaboration patterns in the context of portfolio manager decision-making.

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Abbreviations

AI Artificial Intelligence

API Application Programming Interface

CAPEX Capital Expenditure

CPM Critical Path Method

DIA Data-Intensive Activity

DSR Design Science Research

ERP Enterprise Resource Planning

ESCO European Skills, Competences, Qualifications and Occupations

FHNW Fachhochschule Nordwestschweiz (University of Applied Sciences and Arts Northwestern Switzerland)

GDPR General Data Protection Regulation

KG Knowledge Graph

KIA Knowledge-Intensive Activity

LSTM Long Short-Term Memory

ML Machine Learning

OPEX Operational Expenditure

PPM Project Portfolio Management

RCPSP Resource-Constrained Project Scheduling Problem

SME Small and Medium-sized Enterprise

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A.1 Literature screening sheet columns 33

Appendix A

Literature Screening Sheet

The following columns are used in the Scopus literature screening Excel file:

Table A.1: Literature screening sheet columns

Column	**Description**
Author(s)	Author names
Year	Publication year
Title	Full paper title
Source	Journal / Conference
Relevance	Low / Medium / High — assessed against the four RQs
Finding	Key insight relevant to this research
Category	Which RQ or theme the paper primarily addresses
Notes	Additional remarks

Appendix B

Supervisor Documents

The following documents were provided by supervisor Dr. Emanuele Laurenzi and are available in the `documents/` folder:

- `03-Workshop-Canvas-KI-KMU-EN.pdf` — AI workshop canvas for SMEs (EN)
- `FHNW-KI-KMU-Methodik-Praxisleitfaden.pdf` — FHNW AI methodology guide for SMEs